Activity 2

Let A and B be the set of lines in xy-plane such that A consists of lines parallel to X-axis. For $x \in A$, $y \in B$, let R be a relation from A to B defined by $x\mathrm{R}y$ if x is perpendicular to y. Find the elements of B using a graph sheet.

Illustration 6

Let $A = \{1,3,5,7\}$ and $B = \{4,8\}$. If R is a relation defined by "is less than" from A to B, then 1R4 ($\because$ 1 is less than 4). Similarly, it is observed that 1R8, 3R4, 3R8, 5R8, 7R8

Equivalently R= {(1,4), (1,8), (3,4), (3,8), (5,8), (7,8)}

Note

In the above illustration $A \times B = \{(1,4), (1,8), (3,4), (3,8), (5,4), (5,8), (7,4),(7,8)\}$
R = {(1,4), (1,8), (3,4), (3,8), (5,8), (7,8)} We see that R is a subset of $A \times B$.

Illustration 7

In a particular area of a town, let us consider ten families A, B, C, D, E, F, G, H, I and J with two children. Among these, families B, F, I have two girls; D, G, J have one boy and one girl; the remaining have two boys. Let us define a relation R by $x\mathrm{R}y$, where x denote the number of boys and y denote the family with x number of boys. Represent this situation as a relation through ordered pairs and arrow diagram.

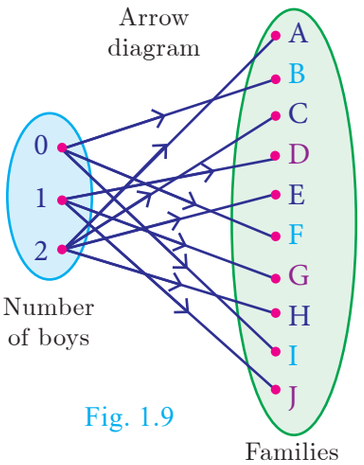


Fig. 1.9

Since the domain of the relation R is concerned about the number of boys, and we are considering families with two children, the domain of R will consist of three elements given by {0,1,2}, where 0, 1, 2 represent the number of boys say no, one, two boys respectively. We note that families with two girls are the ones with no boys. Hence the relation R is given by

$$\mathrm{R} = \{(0,B),(0,F),(0,I),(1,D),(1,G),(1,J),(2,A),(2,C),(2,E),(2,H)\}$$

This relation is shown in an arrow diagram (Fig.1.9).

Example 1.4 Let $A = \{3,4,7,8\}$ and $B = \{1,7,10\}$. Which of the following sets are relations from A to B?

(i) $\mathrm{R}_1 = \{(3,7), (4,7), (7,10), (8,1)\}$ (ii) $\mathrm{R}_2 = \{(3,1), (4,12)\}$

(iii) $\mathrm{R}_3 = \{(3,7), (4,10), (7,7), (7,8), (8,11), (8,7), (8,10)\}$

Solution $A \times B = \{(3,1), (3,7), (3,10), (4,1), (4,7), (4,10), (7,1), (7,7), (7,10), (8,1), (8,7), (8,10)\}$

(i) We note that, $\mathrm{R}_1 \subseteq A \times B$. Thus, R_1 is a relation from A to B.

(ii) Here, $(4,12) \in \mathrm{R}_2$, but $(4,12) \notin A \times B$. So, R_2 is not a relation from A to B.

(iii) Here, $(7,8) \in \mathrm{R}_3$, but $(7,8) \notin A \times B$. So, R_3 is not a relation from A to B.

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